

Program : Diploma in Robotic Process Automation	
Course Code : 5303C	Course Title: Embedded and IOT System Design
Semester : 5	Credits: 4
Course Category: Programme Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To understand the basic concepts and applications of embedded system design
- To understand the concepts of programming an embedded system and basics of Microcontroller
- To understand the fundamentals of IoT and its architectural modelling
- To understand and have thorough knowledge about smart things, functional blocks, communication networks and protocols used in IoT

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Digital Design and Computer Architecture	3302	Digital Design and Computer Architecture	3
Basic functions and features of computer, operating system and internet	1008	Introduction to IT system lab	1

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	To have a thorough understanding about the basic structure and design of an Embedded System	14	Understanding

CO2	To understand the concepts of programming an embedded system and basics of Microcontroller	15	Understanding
CO3	To Understand the IoT fundamentals and architecture modeling	15	Understanding
CO4	To understand the smart things, functional blocks, communication networks and protocols used in IoT	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3	3					
CO3	3	3					2
CO4	3				1		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	To have a thorough understanding about the basic concepts and design of an Embedded System		
M1.01	Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing systems	1	Knowledge
M1.02	Describe the Characteristics and quality attributes of Embedded systems,	2	Understanding
M1.03	Explain the Classification of Embedded systems	1	Understanding
M1.04	Describe about the Core of a typical embedded systems, Memory, Sensor and actuators, Communication Interface, Embedded firmware	3	Understanding
M1.05	Explain the Major application areas of embedded systems	1	Understanding

M1.06	Describe about the Design Process in Embedded System, Design Metrics, Challenges in Embedded System Design,	3	Understanding
M1.07	Explain about Design Process and Design Example of Smart Card	3	Applying

Contents:

Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing systems, Characteristics and quality attributes of Embedded systems, Classification of Embedded systems, **Core of a typical embedded systems**, Memory, Sensor and actuators, Communication Interface, Embedded firmware, Major application areas of embedded systems, **Design Process in Embedded System**, Design Metrics, Challenges in Embedded System Design, Design Process and Design Example of Smart Card

CO2	To understand the concepts of programming an embedded system and basics of a Microcontroller		
M2.01	Programming Embedded Systems- Processor types	1	Understanding.
M2.02	Explain about Selection of programming language, Selection of Operating System	2	Understanding
M2.03	Describe about the Development of Embedded System Software	2	Understanding.
M2.04	Block diagram of 8051, Pin configuration, Registers, Internal Memory,	2	Understanding.
M2.05	Timers: Modes and Applications, Port Structures, Interrupts	2	Applying
M2.06	Assembly Language Programming - Addressing Modes, Instruction set	2	Applying
M2.07	Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display.	3	Applying
M2.08	Interfacing with 8051 using Assembly language programming: LCD display, Keyboard, Stepper Motor	1	Applying
	Series Test – I	1	

Contents:

Programming Embedded Systems- Processor types, Selection of programming language, Selection of Operating System, Development of Embedded System Software.
8051 Microcontroller- Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers: Modes and Applications, Port Structures, Assembly Language Programming - Addressing Modes, Instruction set, Interrupts, Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display.
LCD display, Keyboard, Stepper Motor

CO3	To Understand the IoT fundamentals and architecture modeling		
M3.01	Introduction of IoT, History of IoT	1	Knowledge
M3.02	IoT standards, IoT enabling technologies	1	Understanding
M3.03	IoT physical entities	1	Understanding
M3.04	IoT Building blocks – Architecture	2	Understanding
M3.05	Study of existing IoT platforms	1	Understanding
M3.06	Overview of Wireless sensor Networks	2	Knowledge
M3.07	Overview of RF, WiFi Bluetooth, Zigbee	2	Knowledge
M3.08	Overview of GSM/GPRS	2	Knowledge
M3.09	Requirements of IoT, RFID Basics	1	Understanding
M3.10	WSN topology and deployment strategies	2	Apply

Contents:

Introduction of IoT, History of IoT , IoT standards, IoT enabling technologies, IoT physical entities, IoT Building blocks – Architecture, Study of existing IoT platforms, Overview of Wireless sensor Networks ,Overview of RF, Overview of WiFi ,Bluetooth, Zigbee, Overview of GSM/ GPRS, Requirements of IoT ,RFID Basics ,WSN topology and deployment strategies

CO4	To understand the smart things, functional blocks, communication networks and protocols used in IoT		
M4.01	Introduction of M2M, Evolution of M2M	1	Knowledge
M4.02	Working of M2M, Role of M2M in IoT	2	Understanding
M4.03	IoT communication pattern, Standardization activities towards protocols	2	Understanding
M4.04	TCP/IP Basics, Overview of IPV6	2	Knowledge
M4.05	Network devices & configurations	1	Understanding
M4.06	Overview of Web servers , Socket programming	2	Apply
M4.07	IoT Protocol overview	1	Knowledge

M4.08	Overview of MQTT ,Overview of COAP,6lowpan	2	Understanding
M4.09	Overview of REST, HTTP/HTTPS	1	Understanding
	Series Test – II	1	
Contents: Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT communication pattern, Standardization activities towards protocols, TCP/IP Basics, Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan, Overview of REST, HTTP/HTTPS.			

Text / Reference

T/R	Book Title/Author
T1	Introduction to Embedded Systems by K V Shibu , 2 nd edition, McGraw Hill, 2016
T2	Embedded Systems – Architecture Programming and Design by Raj Kamal, 2 nd edition, McGraw Hill
T3	Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 microcontroller and Embedded System, Pearson Education, 2nd edition.
T4	Vijay Madiseti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1st Edition, VPT, 2014
T5	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017
T6	Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill (India) Private Limited
R1	Embedded Real Time Systems by Dr.K.V.K.K Prasad Dream tech press, 2009.
R2	The 8051 Microcontrollers: Architecture Programming and Applications, K Uma Rao & Andhe Pallavi, Pearson, 2011.
R3	Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things –Key applications and Protocols”, Wiley, 2012
R4	Al-Fuqaha et al. Internet of things: A survey on enabling technologies, protocols, and applications. IEEE Communications Surveys & Tutorials (2015), pp. 2347-2376.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108102169 Introduction to Embedded System Design, IIT Jammu
2	https://nptel.ac.in/courses/117104072 Microcontrollers and applications, IIT Kharagpur
3	https://nptel.ac.in/courses/106105166 Introduction to Internet of Things, IIT Kharagpur